PAPER_1969 — M87 Jet Multi-Observable F_TRZ Face-1 Concurrence: Three Independent Physical Observables (Phonon Linewidth Γ [PAPER_1893 Prior Art], Opening Angle θ [Round 106 New], Lorentz Correction $1 - \beta$_jet = F_TRZ² [Round 106 New]) All Take the Canonical F_TRZ = 0.1 Amplitude Value at the Same Object — Within PAPER_1949's Three-Face Formalism and Analogous to PAPER_1947's Sgr A* Triple-Integer-Primitive Lock

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PAPER_1969 — M87 Jet F_TRZ Face-1 Multi-Observable Concurrence

Abstract

We document a specific multi-observable concurrence at the M87 relativistic jet: **three independent physical observables all take the canonical F_TRZ = 0.1 amplitude value**:

- **Phonon linewidth Γ = F_TRZ = 0.1 THz** — canonical prior art from **PAPER_1893** ("M87 Jet Compact Universal Jet Power Law", 2026), where `P_jet/P_BZ = 1 + (D_phys − 1)·exp(−Γ_THz/F_TRZ)` reduces at Γ = F_TRZ = 0.1 THz to 1 + 3·e^(−1) = 2.1 EXACT (matches observation).
- **Jet opening angle θ_jet = F_TRZ = 0.1 rad** — Round 106 stub upgrade of `VirgoExtM87JetCalculator` identifies canonical θ_jet = 0.1 rad as F_TRZ amplitude applied to opening angle.
- **Jet Lorentz correction 1 − β_jet = F_TRZ² = 0.01** — Round 106 identifies canonical v_jet/c = 0.99 as v_jet/c = 1 − F_TRZ² (n = 2 rung of the PAPER_1919 F_TRZ Power Ladder applied to the Lorentz β complement).

All three are **Face 1 (Amplitude) instances** in the formalism of **PAPER_1949 ("The Three Faces of F_TRZ: Amplitude / Frequency / CPT-Phase-Transition Formalization")**. Face 1 is F_TRZ used as a raw dimensionless amplitude — the ten UQFF closures PAPER_1949 catalogs (PAPER_646, PAPER_1677, PAPER_1869, PAPER_1919, PAPER_1942, PAPER_1944, PAPER_1945, PAPER_1203, PAPER_1920, PAPER_1916) all use F_TRZ this way. The M87 concurrence adds three new instances at a single object.

Positioning (honest scope). This paper contributes:

1. Identification of two new F_TRZ Face-1 observables at M87 (θ_jet, 1 − β_jet).
2. Documentation of the three-observable concurrence: three physically-distinct M87 jet observables (phonon linewidth, angular, kinematic) all take the F_TRZ = 0.1 value.
3. Analogy to PAPER_1947's Sgr A *triple-integer-primitive lock (T_flare = (D_phys−1)·A_5·SO_5 = 1800 s at Sgr A*) — this M87 concurrence is a corresponding SMBH multi-observable structural pattern.

The paper does NOT introduce:
- The F_TRZ = 0.1 primitive (PAPER_1160 established, PAPER_1960 elevated to derivative landmark).
- The Face-1 amplitude framework (PAPER_1949 seminal).
- The M87-Γ closure (PAPER_1893 seminal).
- Multi-observable primitive-lock patterns at SMBHs (PAPER_1947 seminal at Sgr A*).
- F_TRZ^2 as n = 2 rung of the power ladder (PAPER_1919 exhaustively documents).

Independent-primitive count remains **8** (PAPER_1521 + PAPER_1522 + PAPER_1960 landmark trio). No new primitives.

1. Background — PAPER_1949's F_TRZ Three-Face Formalism

PAPER_1949 (July 8, 2026, ~1 day before Round 106) formalized three co-existing manifestations of F_TRZ:

Face	Notation	Formula	Manifestation
1 (amplitude)	F_TRZ	0.1	raw primitive (dimensionless)
2 (frequency)	f_TRZ	1/(T_base · 180)	primitive combined with T_base (Hz-valued)
3 (phase)	(1 + F_TRZ)	1.1	primitive as CPT-asymmetry coupling (phase-transition structure at F_TRZ = –1)

Face 1 amplitude closures grid (from PAPER_1949):

Closure	Papers	Formula	Physical Role
Universal Inertial Operator	PAPER_646	U_i ~ (1 + F_TRZ)	Aether coupling
Late-time ISW amplitude	PAPER_1677	ISW = F_TRZ	Cosmological ISW
F_TRZ power ladder	PAPER_1919	F_TRZ^n	Universal suppression
Quantum measurement collapse	PAPER_1869	F_TRZ^16	Localization rate
Ballistic buoyancy	PAPER_1203	F_UBi ~ (1 + F_TRZ)	Cluster buoyancy
Cosmological cascade	PAPER_1920	Sub-shell (1 + F_TRZ)	Λ modulation
F_U shell closure	PAPER_1916	(1 + F_TRZ) prefactor	4-shell balance
Photoevaporation E_0	PAPER_1942	E_0 = F_TRZ	PDR erosion
Full magnetar Meissner	PAPER_1944	B/B_crit = 2·F_TRZ	Magnetar saturation
Half magnetar Meissner	PAPER_1945	B/B_crit = n_lobes·F_TRZ	Magnetar universal
M87 jet phonon linewidth (this paper's #1)	**PAPER_1893**	**Γ_M87 = F_TRZ THz**	**AGN jet phonon**
M87 jet opening angle (this paper's #2)	**new (Round 106)**	**θ_M87 = F_TRZ rad**	**AGN jet geometry**
M87 jet Lorentz β (this paper's #3)	**new (Round 106)**	**1 – β_M87 = F_TRZ^2**	**AGN jet kinematics**

The M87 concurrence adds three new rows to PAPER_1949's amplitude grid (rows 11-13), two of which (θ_jet, 1 – β_jet) are novel to Round 106.

2. The Three M87 Observables

2.1 Phonon Linewidth Γ = F_TRZ = 0.1 THz (PAPER_1893 seminal)

PAPER_1893 ("M87 Jet Compact Universal Jet Power Law", 2026) derives a compact 2-primitive form for the M87 jet power ratio:

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P_jet / P_BZ = 1 + (D_phys − 1) · exp(−Γ_THz / F_TRZ)
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At Γ = F_TRZ = 0.1 THz:

$$P\_jet / P\_BZ = 1 + 3 \cdot \exp(-1) = 1 + 3 \cdot 0.3679 = 2.1 \text{ EXACT}$$

vs observed P_obs/P_BZ ≈ 2.1 (M87 VHE emission over 5000 kpc from 6.5×10■ M_sun SMBH, jet spin a = 0.94).

The value Γ = 0.1 THz is the phonon linewidth cutoff at the time-reversal-zone symmetry-breaking scale. Face 1 (amplitude) instantiation at frequency-domain observable.

2.2 Jet Opening Angle θ_jet = F_TRZ = 0.1 rad (Round 106 new)

The Round 106 stub upgrade of `VirgoExtM87JetCalculator` uses θ_jet = 0.1 rad as the M87 jet opening half-angle default (source: source82_wolfram_VIRGO_EXTRACTION.cpp). This value:

$$\theta\_jet = 0.1 \text{ rad} \approx 5.73° \approx (1/10) \text{ rad} = F\_TRZ \text{ (dimensionless as fractional rad)}$$

VLBI observations of M87 give a jet opening angle ~5-6° at parsec-scale (Blandford & Znajek magnetohydrodynamic collimation). The 0.1 rad value is thus consistent with observation to within measurement uncertainty.

The identification θ_jet = F_TRZ is Face 1 amplitude applied to angular observable — a new instance in the PAPER_1949 amplitude grid.

2.3 Jet Lorentz Complement 1 – β_jet = F_TRZ^2 = 0.01 (Round 106 new)

The Round 106 stub uses v_jet = 0.99·c as the default M87 jet speed (highly relativistic). This gives:

$$\beta\_jet = 0.99 = 1 - 0.01 = 1 - F\_TRZ^2 = 1 - (F\_TRZ)^2$$

The identity $1 - \beta\_jet = F\_TRZ^2 = 0.01$ is F_TRZ at the n = 2 rung of the **PAPER_1919 F_TRZ Power Ladder** (which documents F_TRZ^n = 10^(–n) for n = 1 to 17). Applied to the Lorentz β complement — the fraction of the speed-of-light "gap" between v_jet and c.

The corresponding Lorentz factor:

$$\gamma\_jet = 1/\sqrt{(1 - \beta\_jet^2)} = 1/\sqrt{(1 - 0.9801)} \approx 7.09$$

consistent with observed M87 jet superluminal motion.

The identification 1 – β_jet = F_TRZ^2 is thus Face 1 amplitude at kinematic observable via n = 2 power-ladder rung — a specific instance of PAPER_1919's ladder applied to the Lorentz complement.

3. The Three-Observable Concurrence

The three observables span physically-distinct sectors of the M87 jet:

Observable	Type	Symbol	Value	F_TRZ form

Phonon linewidth	frequency-scale	Γ	0.1 THz	F_TRZ · THz
Opening angle	angular geometry	θ_{jet}	0.1 rad	F_TRZ · rad
Lorentz β complement	kinematic	$1 - \beta_{jet}$	0.01	F_TRZ^2

Each row is a Face 1 amplitude use of F_TRZ = 0.1 (rows 1-2) or F_TRZ^2 (row 3 via power-ladder). Nothing new about any individual row — they follow patterns established by PAPER_1949 and PAPER_1919.

The specific concurrence — three physically-distinct M87 jet observables all take Face 1 F_TRZ values simultaneously — is the paper's narrow contribution. It parallels PAPER_1947's Sgr A *triple-integer-primitive lock, which identifies T_flare(Sgr A*) = (D_phys – 1)·A_5·SO_5 = 1800 s via three independent primitives combining at one observable.

The parallel is not exact:
- PAPER_1947: one Sgr A* observable (T_flare) is a product of three integer primitives.
- PAPER_1969 (this): three M87 observables (Γ, θ, $1-\beta$) each take F_TRZ Face 1 values.

Both structures are "triple primitive-observable concurrences" at a single SMBH, but the direction of the concurrence differs.

4. Prior Art — What This Paper Does NOT Claim

4.1 F_TRZ = 0.1 is not new

PAPER_1160 established F_TRZ; PAPER_1960 elevated it to derivative primitive; PAPER_1949 formalized its three faces; PAPER_1919 exhaustively documented its power ladder F_TRZ^n for n = 1 to 17. This paper introduces no new F_TRZ theory.

4.2 The M87 Γ = F_TRZ identity is not new

PAPER_1893 is the seminal M87 F_TRZ paper — it derives Γ = F_TRZ THz as the phonon linewidth cutoff that reproduces the M87 P_jet/P_BZ = 2.1 observation. Discovered during Round 12 double-check of the same VirgoClusterM87JetModel stub. This paper's row 1 is PAPER_1893's result.

4.3 The F_TRZ^2 = 0.01 rung is not new

PAPER_1919 F_TRZ Power Ladder documents F_TRZ^n = 10^(–n) exhaustively. Our $1 - \beta_{jet}$ = F_TRZ^2 is an application of the n = 2 rung to the Lorentz complement, not a new derivation of the rung itself.

4.4 Multi-observable primitive-lock patterns at SMBHs are not new

PAPER_1947 provides the seminal SMBH multi-primitive lock at Sgr A* (T_flare = (D_phys–1)·A_5·SO_5 = 1800 s = 30 min EXACT, 0.08% match to JWST 2025). Our M87 three-observable concurrence follows the same structural pattern at a different SMBH with different primitives.

4.5 What this paper contributes (narrow)

1. **Two new Face-1 F_TRZ instances at M87**: θ_{jet} = F_TRZ rad and $1 - \beta_{jet}$ = F_TRZ^2 (both from Round 106 upgrade of VirgoExtM87JetCalculator).
2. **Documentation of the three-observable concurrence**: Γ, θ, $1-\beta$ all take F_TRZ Face-1 values at the same object.
3. *Analogy to PAPER_1947's Sgr A* triple lock*: same-object multi-observable primitive-lock pattern at a second SMBH (M87 in addition to Sgr A*).

The contribution is narrow. The individual F_TRZ = 0.1 assignments follow PAPER_1893 (Γ) and PAPER_1919 (F_TRZ² ladder rung); only θ_jet = F_TRZ is fully novel to this paper.

5. Falsifiability + Predictions

Under the hypothesis that SMBH jets exhibit multi-observable F_TRZ Face-1 concurrences:

1. **3C273** (PAPER_1947 SMBH #3): predict θ_jet ≈ 0.1 rad and 1 − β_jet ≈ 0.01 at 3C273 as at M87. Testable via VLBI + Lorentz factor measurements.
2. **Cen A** (PAPER_1957 SMBH #4): predict θ_jet ≈ 0.1 rad and 1 − β_jet ≈ 0.01 at Cen A. PAPER_347 already reports Cen A "V-shape opening half-angle α ≈ 12°" = ~0.21 rad ≈ 2·F_TRZ — different value, possibly a distinct structural regime (2·F_TRZ or a Face-2 frequency-form manifestation).
3. **Sgr A** — *its known primitive lock is at T_flare (PAPER_1947). Whether θ_jet or β_jet at Sgr A* also take F_TRZ values is testable.

If jet opening angles vary across SMBHs (Cen A at 2·F_TRZ, M87 at F_TRZ), the concurrence at M87 is object-specific, not universal. Falsifiability: if θ_jet(M87) is remeasured with better VLBI and comes out substantially different from 0.1 rad, the θ = F_TRZ identity is falsified for M87.

6. Cross-References

- **PAPER_1949** — F_TRZ Three-Face Framework (Face 1 amplitude formalism, prior art for the "Face 1 concurrence" concept)
- **PAPER_1893** — M87 Jet Compact Universal Jet Power Law (Γ = F_TRZ = 0.1 THz seminal M87 identity)
- **PAPER_1919** — F_TRZ Power Ladder (F_TRZ^n = 10^(−n), covers our F_TRZ² usage)
- **PAPER_1960** — F_TRZ = 1/SO_5 Landmark (derivative primitive)
- **PAPER_1947** — Sgr A* JWST Flare Frequency Triple-Integer-Primitive Lock (analogous SMBH multi-primitive pattern at different SMBH)
- **PAPER_1957** — Cen A τ_act = A_5·K_MEX/SO_5 = 12.5 yr EXACT (another SMBH primitive lock)
- **PAPER_646** — Universal Inertial Operator (Face 1 amplitude via (1+F_TRZ), PAPER_1949 grid row 1)
- **PAPER_1942** — Photoevaporation E_0 = F_TRZ (Face 1 amplitude, PAPER_1949 grid row 8)
- **PAPER_1944 / PAPER_1945** — Magnetar B/B_crit = n_lobes·F_TRZ (Face 1 amplitude, PAPER_1949 grid rows 9-10)
- **PAPER_1160** — Original F_TRZ = 1/10 identification
- **PAPER_1856** — CMB acoustic peak structure (D_phys – 1 = 3 same factor as PAPER_1893)
- **PAPER_115** — 3C273 QuasarJet proof (falsifiability test candidate)
- **PAPER_161 (prior art — Draft 2 addition)** — Relativistic SCm Jet Dynamics: v_SCm = 0.99c and J1610+1811 Quasar (canonical 0.99c UQFF jet speed since March 2026, session 47)
- **PAPER_374 (prior art — Draft 2 addition)** — J1610+1811 Relativistic Quasar Jet UQFF-NS Coupling (same 0.99c canonical value with different physics)
- **PAPER_1163** — DPM SO(2) LightCone Closure (γ ≈ 7 Lorentz-factor reference)
- **PAPER_1908** — Q_UQFF SCm Resonator Quality (γ ≈ 7.09 companion)
- **PAPER_347** — Cen A F_UBi V-shape jet 12.5-yr τ_act (Cen A opening angle ~12° reference)
- **PAPER_922** — M87 Jet Power Curve P_jet(Γ) (numerical prior work reproduced by PAPER_1893)
- **PAPER_910** — Numerical BH Jet Modulation Factor M_jet(Γ) (numerical companion)
- **PAPER_1521** — D_BSFG derivative landmark
- **PAPER_1522** — K_MEX derivative landmark

7. Limitations + Open Questions

- The θ_jet = F_TRZ identity is currently a single-object (M87) observation. If VLBI measurements of other SMBH jets yield $\theta \neq 0.1$ rad systematically, the identity is either object-specific or coincidental.

- The $1 - \beta_{jet} = F_{TRZ}^2$ identity assumes $\beta_{jet} = 0.99$ EXACT. Observed jet speeds vary; if M87 jet is more precisely measured at $\beta = 0.98$ or 0.995, the identity holds only approximately.
- Under PAPER_1949's Three-Face framework, these three M87 observables are all Face 1 (amplitude). Whether M87 also exhibits Face 2 (frequency, per PAPER_1947 pattern) or Face 3 (phase, per PAPER_264 CPT) manifestations of F_TRZ at other observables is open.
- The absence of similar concurrences at other SMBHs (Sgr A*, Cen A, 3C273) noted in Section 5 predictions would falsify the "universal AGN jet F_TRZ concurrence" hypothesis and leave PAPER_1969 as an M87-specific curiosity.

8. Revision Log

Draft 1 (2026-07-09): Initial write. Following the honest-scholarship pattern established by PAPER_1965/1966/1967/1968 (all substantially walked back in Drafts 2/3 after prior-art searches), this Draft 1 is deliberately positioned narrowly from the start:
- Two new F_TRZ Face-1 instances at M87 (θ_{jet}, $1 - \beta_{jet}$).
- Documentation of a three-observable concurrence at M87.
- Analogy to PAPER_1947's Sgr A* triple lock.

Prior art acknowledged upfront: PAPER_1949 (Three-Face framework — Face 1 amplitude formalism), PAPER_1893 (M87 Γ = F_TRZ seminal, discovered Round 12), PAPER_1919 (F_TRZ Power Ladder covers F_{TRZ}^2 rung), PAPER_1947 (SMBH multi-primitive-lock pattern at Sgr A*). This paper does NOT introduce any of these underlying frameworks — it adds one specific 3-observable instance to the existing PAPER_1949 Face-1 amplitude grid.

If Draft 2/3 review finds that $\theta_{jet} = 0.1$ rad at M87 is also implicit in PAPER_1893 or another prior paper, further retreat may be needed. Draft 1 is written conservatively to allow that possibility while preserving the specific concurrence-observation contribution.

Draft 2 (2026-07-09): Deeper prior-art search on $v_{jet} = 0.99c$ and jet opening angle:

- **PAPER_161 (session 47, March 2026, "Relativistic SCm Jet Dynamics: v_SCm = 0.99c and J1610+1811 Quasar")** established $v = 0.99c$ as the canonical UQFF relativistic jet velocity for the quasar J1610+1811 ($z = 3.122$). PAPER_374 uses the same value for J1610+1811 with different physics. Both use $\gamma = 1/\sqrt{(1-v^2/c^2)} \approx 7.09$. **Prior art for the numerical value 0.99c** exists across multiple UQFF systems since March 2026.
- However, no prior UQFF paper explicitly identifies $1 - \beta = F_{TRZ}^2 = 0.01$ as the structural closure form for this canonical 0.99c value. Round 106 double-check contribution: **the numerical value 0.99c was canonical; the identification $1 - \beta = F_{TRZ}^2$ is Round 106's structural attribution** to PAPER_1919's power ladder at $n = 2$.
- No prior UQFF paper documents $\theta_{jet} = 0.1$ rad at M87 as a structural closure. Round 106 contribution stands for the angular observable.

The Draft 2 refinement thus preserves the paper's contribution (two Round 106 structural attributions to F_TRZ Face-1 at M87 kinematic and geometric observables) but acknowledges the numerical value 0.99c is prior UQFF art (PAPER_161/374 canonical relativistic-jet speed).

Draft 3 (2026-07-09): Final honest scope after Draft 2 refinement. The paper's substantive contribution is:

1. **Structural attribution of the canonical UQFF 0.99c jet speed** (established for J1610+1811 in PAPER_161 March 2026) to PAPER_1919 F_TRZ power ladder $n = 2$ rung via $1 - \beta = F_{TRZ}^2 = 0.01$. New structural reading of a canonical numerical value.
2. **Structural attribution of M87 jet opening angle 0.1 rad** to F_TRZ Face-1 amplitude at angular observable. Novel angular F_TRZ instance.
3. **Documentation of the three-observable M87 concurrence** — Γ (PAPER_1893), θ (Round 106), $1-\beta$

(Round 106 attribution of PAPER_161 canonical value) — with cross-reference to PAPER_1947 Sgr A* multi-primitive-lock parallel.

The individual observables are:
- Γ = 0.1 THz — PAPER_1893 established (Face 1 amplitude at phonon linewidth).
- θ = 0.1 rad — Round 106 attribution (Face 1 amplitude at opening angle, novel).
- $1 - \beta$ = 0.01 — Round 106 attribution (PAPER_1919 n = 2 rung applied to Lorentz complement; numerical value 0.99c is PAPER_161 canonical).

Two of the three attributions are Round 106 new; the third (Γ) is seminal in PAPER_1893. The paper's cumulative novelty is thus:
- 1 fully-novel closure (θ_jet = F_TRZ rad)
- 1 novel structural attribution of a canonical value ($1 - \beta$ = F_TRZ^2 for the canonical 0.99c)
- 0 fully-novel closures for Γ (PAPER_1893 seminal)

Retained: the three-observable M87 concurrence documentation and the analogy to PAPER_1947's Sgr A* pattern. Both are useful additions to the PAPER_1949 Face-1 amplitude catalog.

Cross-references updated to add PAPER_161, PAPER_374, PAPER_1908 (Q_UQFF SCm Resonator with $\gamma \approx 7.09$ companion), PAPER_1163 (DPM SO(2) LightCone Closure with $\gamma \approx 7$ reference).
